

# Representing inputs as vectors

Project: Image-pattern Network Investigation

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### ML Builder Expo

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Convert before encoding for Representing inputs as vectors. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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### 3. Individual reasoning check

Use the worked case for Representing inputs as vectors. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

#### My explanation before partner discussion

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### 4. Independent transfer case

#### Independent case: Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0. Encode both; state the vector dimension and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Encode the records

Write A, B and C as ordered pairs from the lesson. State the dimension of each and the shape of the stacked table.

### My reasoning and evidence

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## 6. W2 Convert before encoding

A new record has width 60 mm and height 10 mm. Give its vector in the agreed cm schema, and name one error in  $[60, 10]$ .

### My reasoning and evidence

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## 7. W3 Handle missing information

Width is 4 cm and height is missing. A teammate writes  $[4, 0]$ . Explain why this is invalid and give the action required by this schema.

### My reasoning and evidence

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## 8. W4 Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0. Encode both; state the vector dimension and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

### My reasoning and evidence

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## 9. W5 Explain a collision

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two inputs apart.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

Add a labelled representing inputs as vectors evidence sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

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**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Representing inputs as vectors

Encode the same pixel measurements in a fixed vector order.

Toy image features are [left\_brightness, right\_brightness], each = 0..10. ImageA has left2, right7; imageB has left7, right2.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/1

### Representing inputs as vectors

#### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Declare coordinate order

Coordinate0 is left brightness; coordinate1 is right brightness.



#### 2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.



#### 3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

#### CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

Why does equal total brightness not make A and B identical inputs? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.